

[Green Software Engineering] Survey – Perceptions and Practices in the Software Industry

Hello! I hope you are doing well!

First of all, thank you for being here and contributing to this research. The objective of our study is to investigate the adoption of sustainable practices in software development, with a focus on Green Software Engineering. This topic is essential to understanding how the technology sector can contribute to environmental sustainability throughout the software life cycle.

This research is divided into two sections:

Section 1 (Current): In this first stage, we present the research objectives, the terms for participation, possible risks and benefits, and we collect your free and informed consent.

Section 2: Next, you will be asked to complete a form divided into three parts: (1) a brief professional profile survey, (2) an assessment of organizational maturity in the adoption of sustainable practices, and (3) a set of open-ended questions about your perception and experience with the topic.

The estimated time to complete the entire form is approximately 10 to 15 minutes.

TECHNICAL INFORMATION

This research was approved by the Research Ethics Committee for Human Beings of the Federal University of Alagoas (UFAL), in accordance with all applicable ethical standards and guidelines.

Research title: Evidence of the adoption of Green Software Engineering practices: an investigation of the state of the art and practice in the software industry

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** Indica uma pergunta obrigatória*

1. All data collected here are stored under the principle of anonymity to ensure the safety of all volunteers. To proceed with the research, please indicate your agreement with the Free and Informed Consent Form and enter your full name below. *

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☐ I agree

Assessing Sustainable Practices in the Software Life Cycle According to the Green Software Foundation Guidelines

2. Full Name: *

3. For which type of application or platform is/are the project(s) you currently work on being developed? *

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☐ Mobile Applications

☐ Web

☐ Desktop

☐ Embedded Systems

☐ IoT

☐ APIs

☐ Cloud Platform

☐ Outro:

Area 1 – Control of High Energy-Consuming Functionalities

4. 1 – Does the software project you participate in identify the functionalities that consume the most energy? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

5. 2 – Are functionalities that require more computational resources prioritized for review and optimization in your current project? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

6. 3 – Are tools used in the project to map and monitor the energy consumption of functionalities? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

7. 4 – Do design decisions take into account the energy impact of the most frequently used functionalities? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

8. 5 – Is the control of high energy-consuming functionalities treated as a priority in the project you collaborate on? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

Area 2 – Reduction in Data Usage

9. 6 – Are there guidelines to reduce data usage in the organization's systems (e.g., minimizing the volume of data transmitted, stored, or processed)? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

10. 7 – Are practices such as compression, caching, and media reduction adopted on a recurring basis? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

11. 8 – Is data usage in the project optimized with the goal of reducing the use of computational resources? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

12. 9 – Is the team trained to apply data-saving strategies? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

13. 10 – Is data usage reduction integrated into the software development process? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

Area 3 – Removal or Refactoring of Inactive Functionalities

14. 11 – Is there an active policy to identify obsolete or unused functionalities? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

15. 12 – Is inactive code regularly removed or refactored? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

16. 13 – Are tools used to detect and eliminate dead code? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

17. 14 – Are refactorings planned to make the software leaner and more efficient? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

18. 15 – Is code cleanup a recurring practice in software deliveries? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

Area 4 – Elimination of Unnecessary Loops and Redundant Processing

19. 16 – Are inefficient or unnecessary loops identified during development? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

20. 17 – Are unnecessary processing tasks avoided through good coding practices? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

21. 18 – Does the project use code analysis tools to detect inefficient patterns? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

22. 19 – Does the team understand the energy impacts caused by poorly designed loops? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

23. 20 – Is there concern about avoiding unnecessary energy consumption during system execution? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

Area 5 – Adaptation to Energy Mode and Device Conditions

24. 21 – Does the behavior of the developed systems adapt according to the energy mode of the execution environment? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

25. 22 – Are system resources automatically reduced in low-energy situations? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

26. 23 – Does the developed software respond to contextual conditions, such as connectivity or battery usage? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

27. 24 – Is context adaptation considered from the design phase? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

28. 25 – Are systems designed to adapt their behavior according to the operational conditions of the execution environment? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

Area 6 – Limiting Computational Precision According to Need

29. 26 – Are levels of precision applied in accordance with the system's functional requirements? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

30. 27 – Is the resolution or precision of calculations reduced when they are not essential? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

31. 28 – Are there formal guidelines to limit unnecessary complex calculations? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

32. 29 – Are systems developed considering the balance between precision and the use of computational resources? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

33. 30 – Is this practice known and applied by the project's developers and architects? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

Area 7 – Real-Time Monitoring of Energy Consumption

34. 31 – Is the energy consumption of the systems monitored in real time? *

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- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

35. 32 – Are indicators defined to monitor the energy impact per module or component? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

36. 33 – Are architectural decisions guided by energy consumption metrics? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

37. 34 – Are tools used to collect energy metrics during execution? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

38. 35 – Is energy consumption monitored with the same rigor as performance and security? *

Marcar apenas uma oval.

- ☐ 1 – No
- ☐ 2 – Slightly
- ☐ 3 – Partially
- ☐ 4 – Almost Totally
- ☐ 5 – Totally

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